Tester Present

Specialist Automotive Solutions

TESTER ENGINEERING SUITE · V1.0.9-2026

TESTER ENGINEERING SUITE

USER MANUAL

User manual & operator's guide — from first connection to full flash.

"Engineered like infrastructure, not a scan tool."

EDITION v1.0.9-2026 · Windows 7+ (x86 / x64)
APPLIES TO Light-vehicle platform + Heavy overview
WEB tester.engineering

Right to Repair

BUILT BY A MECHANIC, FOR THE RIGHT TO REPAIR

BEFORE YOU START

HOW TO READ THIS MANUAL

This manual walks the Tester Engineering Suite in the order you actually use it: connect, identify, read, diagnose, then — only when you mean to — write. Each workspace lives on its own tab in the main window; this guide tells you which tab, what it does, and where the sharp edges are.

You do not have to read cover-to-cover. Jump to the section you need. But if you are new to the suite, read "Getting started" and "Connecting to a vehicle" once — everything else assumes a live, correctly-selected connection.

Safety symbols used throughout

NOTE **Note**

Context you want before running the feature — where it lives, what it assumes, how to read the result.

TIP **Tip**

A faster or cleaner way to do the job, or a setting worth changing once.

WARNING **Warning**

An irreversible or vehicle-affecting action. Read it before you click. These are the operations that clear monitors, lock modules, or write to an ECU.

The three rules that never change

1. The suite is live-fire by default. Transmit-capable features are armed when you open them — this is a professional tool, not a toy. The single confirmation prompts on genuinely irreversible actions (flash, memory write, DTC clear, security access) are there on purpose; do not treat them as noise.
2. Restraint and airbag modules are hard-blocked from routine and output-state commands, at both the menu and the executor. Reads are allowed; actuation is not. This cannot be overridden.
3. Your data stays yours. The suite collects nothing about the vehicles you work on, and there is a hard wall between vehicle data and licensing. Licence checks never see VINs; vehicle work never touches the licence server.

01 GETTING STARTED

Install, licence, and pick your interface.

INSTALLING

4. Run the installer and launch "Tester Engineering Suite v1.0.4-2026". The suite is a Windows desktop application; no service or driver of its own is installed.
5. Install your interface's own J2534 PassThru drivers if you have not already (Tactrix, Drew Technologies, OBDX Pro, VCM2, etc.). The suite finds any correctly-installed PassThru DLL automatically.
6. On first launch you are asked to activate a licence (below). Until then the suite runs in a limited state.

ACTIVATING YOUR LICENCE

Licences are bound to your machine fingerprint. Lifetime tiers ship with a USB hardware token — leave it plugged in.

7. Enter the licence key and the name/email it was issued to. The suite validates online once, then caches the result (encrypted, per-user) so you can work offline afterwards.
8. If you move to a new PC, request a re-issue — the fingerprint will differ. Heavy-duty runs fully offline against its USB key and never contacts a server.

> **NOTE** **Your licence never sees a VIN**
>
> Licence validation and vehicle diagnostics are separated by design. The licence path carries your key, name and email — never anything read off a car.

CHOOSING AN INTERFACE

Any SAE J2534 PassThru device works. ELM327 adapters are supported for basic OBD-II only — the suite will tell you when a task needs a real pass-through device (flash and security access are not possible over ELM327).

02 THE MAIN WINDOW

One window, one connection, many workspaces.

The suite is organised as a single window with a menu strip along the top, a status bar along the bottom, and a large tabbed work area in the middle. When you decode a VIN, the workspace auto-themes to that manufacturer – the banner and accent colour change so you always know which vehicle you are on.

Menu strip

MENU	WHAT LIVES THERE
File	Open/save binaries and logs, export reports, exit.
Tools	The feature launchers that are not full tabs – utilities, wizards and settings.
Help	About, version/build info, and links back to tester.engineering.
Developer (hidden)	Ctrl+Shift+E – suite self-diagnostics. See the Power-User Addendum.
Advanced Diagnostics (hidden)	Ctrl+Shift+A – 30+ engineering tools. See the Power-User Addendum.

Status bar

The bottom bar is your source of truth for connection state: transport (J2534 / ELM327 / Remote), protocol, TX/RX IDs, battery voltage, live TX/RX frame counters, current VIN and DTC count. If something isn't working, read the status bar first – it usually tells you why (NO TRANSPORT, OFFLINE, COMM FAILURE).

Workspaces (tabs)

Tabs are grouped by job: Connection, Diagnostics, Live Data, Diagnostic Services, ECU Flashing, Calibration, EEPROM, Emissions, Bus tools, Crash Data, Heavy-vehicle datalinks, plus per-vehicle workspaces that appear when a matching VIN is read. Some tabs are VIN-gated and stay hidden until relevant.

03 CONNECTING TO A VEHICLE

There is one connection, and it is global.

> **TIP** **One connection, everywhere**
>
> You connect once, on the Connection tab. Every other workspace uses that same session — there is no per-tab device picker to keep in sync. Change the connection in one place and the whole suite follows.

THE CONNECTION SEQUENCE

9. Open the Connection tab. Pick your interface from the discovered device list. If nothing is listed, confirm the PassThru drivers are installed and the adapter is plugged in.
10. Plug into the OBD-II port (or your bench harness) and turn the ignition on. Click Connect. The status bar moves to Connecting… then Connected.
11. Let the suite auto-detect the protocol, or force one for legacy vehicles (ISO 9141, KWP2000, J1850). CAN-FD and DoIP are detected automatically where present.
12. Read the VIN. This sets the vehicle for the whole session, applies OEM theming, and unlocks any vehicle-specific tabs.
13. Run a module scan to discover which ECUs answer and on which bus.

> **WARNING** **Right bus, right module**
>
> Some vehicles run more than one CAN bus at different speeds (e.g. a high-speed powertrain bus and a medium-speed body bus). Probing a module that lives on the other bus can push the interface toward bus-off. The topographer and module scan show you which bus owns each module — trust them before sending traffic.

IF THE BUS LOOKS DEAD

A sticky "CAN unresponsive" decision can persist after a bad first probe. If you have fixed the wiring/ignition and it still won't talk, reset that flag from the Developer menu (Ctrl+Shift+E → Session & State → Reset CAN-Known-Dead Flag) and re-probe from scratch.

04 READING THE VEHICLE

Identify, scan, and pull faults — all non-destructive.

VIN & VEHICLE IDENTITY

The VIN read and offline VIN decoder break the number into WMI / VDS / VIS and attribute make, model family and build details. Use the decoder (also in Advanced Diagnostics → Identification → Offline VIN Decoder) to sanity-check a VIN even without a connection.

MODULE SCAN & TOPOGRAPHER

The module scan lists every ECU that responds, its addresses and protocol. The CAN & K-Line topographer goes further — it maps the network and animates which modules sit on which bus. Probing is read-only and only adds a node when a module actually answers, so the map reflects reality, not guesses.

FAULT CODES (DTCS)

The DTC workspaces read stored, pending and permanent codes (OBD-II Modes 03/07/0A and UDS service 0x19). Each code is decoded to its SAE J2012 family with a severity hint; where an OEM-specific definition is licensed, you get the manufacturer text and a triage walkthrough.

- **Freeze-frame —** the captured conditions at the moment a code set (Mode 02 / snapshot records).
- **Snapshot / diff —** capture codes before a repair and again after, and compare (Advanced Diagnostics → Fault Diagnosis → DTC Snapshot / Diff).

> WARNING **Clearing codes is not a repair**
>
> Clearing DTCs (service 0x14 / Mode 04) also resets emissions readiness monitors — the car will fail an inspection until a full drive cycle re-runs them. The suite asks before it clears. Fix the fault, then clear, then re-scan to confirm it stays gone.

READINESS & ON-BOARD TESTS

The I/M Readiness Monitors tab shows monitor completion so you know whether a car is ready for an emissions test. The Mode 06 browser exposes on-board test results (660 test definitions bundled) for the sensors and systems that do not throw a code until they fail outright.

05 LIVE DATA, LOGGING & SCOPE

Watch the vehicle work, then prove what you saw.

LIVE PIDS & DATALOGGING

Pick from 1000+ PID definitions or build a custom PID for anything not in the library. The live datalogger plots multiple channels together, records to disk, and plays back with a scrubber and histogram view. Dashboards give you gauges for the drive; the signal analyser gives you waveforms for the diagnosis.

USB OSCILLOSCOPE

Bring an oscilloscope into the same timeline as the CAN data: PicoScope, SCPI/LXI bench scopes and the Hantek 6022 are supported. Scope traces and J2534 PIDs are logged together, so an electrical event and its data-bus consequence line up on one screen.

ENGINE HEALTH WITHOUT A TEARDOWN

- **Relative compression —** from crank-sensor ripple, ranks cylinder contribution (Advanced Diagnostics → Engine Diagnostics).
- **Misfire counter (live) —** per-cylinder misfire counts as they happen.
- **Fuel-trim & O2 tools —** long-term fuel-trim trender and O2 cross-count analyser for mixture and sensor faults.
- **Acoustic diagnostics & NVH —** FFT spectrogram and order-tracking cursors turn a microphone into a noise/vibration finder.

06 DIAGNOSTIC SERVICES (UDS)

The full ISO 14229-1 service surface — with guard rails.

The Diagnostic Services tabs expose UDS directly: session control (0x10), ECU reset (0x11), read/clear DTC (0x19/0x14/0x85), read/write data by identifier (0x22/0x2E), routine control (0x31), I/O control (0x2F), communication control (0x28), security access (0x27) and memory read/write. A raw / custom Mode 22 tab lets you request any DID.

Sessions & keep-alive

Many services only work inside an extended or programming session. Enter the right session first; the suite keeps the session alive with periodic tester-present so a long operation does not time out.

Security access (0x27)

> **WARNING** **Failed unlocks can lock you out**
>
> Most modules count failed seed/key attempts in non-volatile memory and impose a lockout that survives ignition cycles. The suite asks before a security-access attempt on a live vehicle. Be sure you have the right algorithm and target before you try.

The Advanced Diagnostics → Reverse Engineering → Seed-Key Sandbox lets you work out and verify a key against a known seed offline, before you spend an attempt on the car.

Routine & I/O control

Routine control (0x31) runs service routines; I/O control (0x2F) drives actuators and forces outputs. Both are gated by the safety router: they are refused outright for restraint/airbag modules, and confirmed for anything that moves. If a routine or output could fire something dangerous, treat the confirm prompt as the last check before it happens.

07 PROGRAMMING & FLASH

The most powerful — and most dangerous — part of the suite.

The ECU Flashing workspace reprograms supported modules using validated flash profiles. It is built to fail safe: it validates before it writes, watches the battery the whole time, and verifies afterwards.

A SAFE FLASH, STEP BY STEP

14. Confirm the vehicle and target module. The suite will not let you load a profile that does not match the identified vehicle without an explicit override.
15. Load the calibration/firmware file. If it needs a checksum correction, the suite flags it and can correct it before write (CRC / Fletcher).
16. Run the dry-run validation. This exercises the whole sequence — session, security, block layout — without writing a byte. Read the result. A dry-run that is not clean is a flash that will not finish.
17. Connect a stable power supply. Battery voltage is supervised throughout; a sag mid-write is the classic way to brick a module.
18. Arm and run the live flash. Progress, block-by-block, is shown; the operation can abort and roll back on error.
19. Let the post-flash verification finish — calibration-ID / CVN read-back confirms the module took the new image.

> **WARNING** **Battery, dry-run, and never walk away**
>
> A flash is the one operation where a shortcut can cost a module. Always dry-run first, always run from a supply, and never leave a live write unattended. Restraint/airbag modules are not flashable through this workspace at all.

CONFIGURATION & SERVICE TOOLING

- **As-built / configuration editor —** read a module's configuration block, edit fields, and write it back with change tracking.
- **Guided routine runner —** run service routines (resets, calibrations, learns) with plain-language descriptions instead of raw routine IDs.
- **Immobiliser / PATS wizard —** guided anti-theft and key workflows with live security-state monitoring; a held security session keeps the grant alive across the procedure.
- **ODX → routine importer —** import standardised diagnostic descriptions and turn them into runnable routines.

08 CALIBRATION, MEMORY & CRASH DATA

Bench work: maps, EEPROM, clusters and event records.

CALIBRATION EDITOR (A2L / XDF)

Open a definition (ASAM MCD-2 MC A2L or TunerPro XDF) against a binary and edit maps directly: live heatmap, 3-D surface view and map math. Checksum correction and safe, reversible write-back are built in. Emissions-relevant tables are flagged and risky edits are double-confirmed.

EEPROM & CLUSTER TOOLING

> **NOTE** **EEPROM work never touches the vehicle bus**
>
> The EEPROM workspace is strictly a bench tool for chips read off the module. It is deliberately isolated – it will not transmit on the car. Keep it that way.

The EEPROM editor and cluster tooling cover a large library of modules and clusters for odometer-correct repairs and module replacement on the bench. Cluster graphics/flash tooling handles supported instrument clusters.

CRASH DATA (EDR / CDR)

The Crash Data workspace reads restraint/airbag event records and produces a signed, tamper-evident PDF suitable for a file. Identification is deterministic: the VIN selects the make, the make selects the restraint module's fixed address – the suite never scans blindly near a pyrotechnic module. Reads only; the workspace issues no actuation.

09 EMISSIONS & BUS TOOLS

Compliance lanes and the instruments that read the wire.

EMISSIONS

- **I/M readiness —** monitor-completion view so you know a car is ready to test.
- **IM240 / RG240 test lane —** transient-cycle emissions lane with compliant mass-emission calculation and a driver's-aid trace for holding the cycle.
- **Emissions snapshot —** capture the emissions-relevant state for the record.

BUS & TRANSPORT TOOLS

- **Multi-bus CAN sniffer —** capture HS/MS/SW-CAN with filtering, replay and per-ID hints; catches the slow-burn faults a single-shot test misses.
- **CAN bit-timing analyser —** sample-point and timing checks for a bus that works sometimes.
- **CAN bus health check (J2284-3) —** single-shot termination / dominant-recessive / error-counter classification; the right first move when U-codes are present.
- **Filters —** define and manage acceptance filters for focused capture.

10 HEAVY-DUTY & MOBILE PLANT

Trucks, buses and plant – offline, always.

Where a heavy interface (RP1210) is selected, the heavy-vehicle workspaces appear. These cover the datalinks the light-vehicle world never sees. The heavy edition is offline-first: no phone-home, USB-key licence, everything works with the network cable pulled.

- **SAE J1939 –** DM1–DM19 fault handling, module discovery, live SPN/FMI decoding.
- **Caterpillar Data Link (CDL) –** CAT CDL monitoring and diagnostics via supported adapters.
- **J1587 / J1708 –** legacy pre-2007 heavy-vehicle diagnostics.
- **UDS-on-J1939 –** modern UDS services over J1939 for Isuzu, Hino, Fuso and Paccar-family modules.
- **RP1210 adapters –** Nexiq, Dearborn, DrewLinQ and Jaltest interfaces.

> **TIP** **You can force the heavy tabs on the bench**
>
> No truck adapter to hand? Developer menu (Ctrl+Shift+E) → UI Diagnostics → Heavy-vehicle workspaces → "Always show" makes the heavy tabs visible for bench work.

11 SHOP WORKFLOW & RECORDS

The paperwork side, done properly.

- **VIN intelligence brief –** registration, recall and complaint intelligence from Australian and NHTSA sources. The suite is explicit about what a plate lookup can and cannot return.
- **Per-VIN technician notes –** notes are kept per vehicle and superseded, never silently deleted, so the history is always intact.
- **Service schedule & maintenance –** an interval library and per-vehicle maintenance planning.
- **Work orders, labour timer & reports –** build a job, time the labour, and export a customer-facing PDF.
- **Pre / post scan & DVI –** insurer-ready before/after scans and a digital vehicle inspection.

> **NOTE** **Reports you can stand behind**
>
> Exported reports carry a tamper-evident manifest – a cryptographic seal over the content – so a customer or insurer can confirm the report has not been altered since you produced it.

12 INTELLIGENCE & REMOTE

An advisor over the tools, and a bench you can drive from afar.

DIAGNOSTIC INTELLIGENCE

The Diagnostic Intelligence workspace is an advisory layer that reads across the tools — codes, live data, freeze frames — and proposes a ranked set of likely causes and next tests as a branching reasoning tree, with failure treated as a first-class outcome. It proposes; you confirm. It never actuates on its own, and it hard-blocks anything that would touch a restraint module.

REMOTE MODE (MASTER / SLAVE)

Remote mode lets one machine (the master) drive a shop bench (the slave agent) over an encrypted WebSocket link. Diagnostics, security and flash flows are mirrored so a remote session behaves the same as a local one. Pair the two ends, and the master's workspaces operate the remote vehicle as if it were plugged in locally.

13 REFERENCE

Shortcuts, safe-operation rules, and where to get help.

KEYBOARD SHORTCUTS

CHORD	ACTION
Ctrl + Shift + A	Toggle the hidden Advanced Diagnostics menu (30+ engineering tools).
Ctrl + Shift + E	Toggle the hidden Developer menu (suite self-diagnostics).

Both hidden menus are documented in full in the Power-User Addendum. They do not persist — relaunching the suite hides them again.

SAFE-OPERATION CHECKLIST

- Read the status bar before blaming the vehicle — transport, protocol and voltage live there.
- Confirm which bus owns a module before you send traffic to it.
- Dry-run every flash; run from a power supply; never walk away from a live write.
- Ask the customer before you clear codes or attempt security access on a live car.
- Restraints/airbags are read-only — the suite enforces it, and so should you.

TROUBLESHOOTING QUICK HITS

SYMPTOM	FIRST THING TO CHECK
No devices listed	PassThru drivers installed? Adapter plugged in? Try List J2534 Devices (Ctrl+Shift+E → Connection).

SYMPTOM	FIRST THING TO CHECK
Connects, no data	Right protocol and right bus for that module? Ignition on?
Bus won't respond after a fix	Reset the CAN-Known-Dead flag (Ctrl+Shift+E → Session & State).
Flash won't complete	Re-run the dry-run; check battery voltage and target match.
Feature greyed out	It may be VIN-gated or need a real J2534 device (not ELM327).

GETTING HELP

Documentation, licence requests and support: tester.engineering. Keep the Power-User Addendum alongside this manual for the advanced and developer menus.

> **TIP** **Built by a mechanic, for the right to repair**
>
> If a workflow fights you or a description here does not match what you see, tell us — this tool is judged by one question: did it make the job easier for the mechanic?